

Kinematics

ANSWER KEY



Displacement, velocity, and acceleration

- 1 A particle moves with displacement $s = t^3 - 6t^2 + 9t$ (in meters, t in seconds). Find expressions for its velocity v and acceleration a .
 $v = \frac{ds}{dt} = 3t^2 - 12t + 9$. $a = \frac{dv}{dt} = 6t - 12$.
- 2 A particle moves with displacement $s = 2t^3 - 9t^2 + 12t$ (in meters, t in seconds). Find its velocity at $t = 2$ seconds.
 $v = \frac{ds}{dt} = 6t^2 - 18t + 12$. At $t = 2$: $v = 6(4) - 18(2) + 12 = 24 - 36 + 12 = 0$ m/s.

The suvat equations for constant acceleration

- 3 A car starts from rest and accelerates uniformly at 2.5 m/s^2 for 8 seconds. Find its final velocity and the distance traveled, using $v = u + at$ and $s = ut + \frac{1}{2}at^2$.
 $v = u + at = 0 + 2.5(8) = 20$ m/s. $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2.5)(64) = 80$ m.
- 4 A ball is thrown upward with initial velocity $u = 15$ m/s. Using $g = 9.8 \text{ m/s}^2$ (deceleration), find the maximum height reached, using $v^2 = u^2 + 2as$ with $v = 0$ at the top.
 $0 = 15^2 + 2(-9.8)s$, so $s = \frac{225}{19.6} \approx 11.48$ m.
- 5 A cyclist decelerates uniformly from 12 m/s to rest over a distance of 36 meters. Find the deceleration, using $v^2 = u^2 + 2as$.
 $v^2 = u^2 + 2as$: $0 = 12^2 + 2a(36) = 144 + 72a$, so $a = -2 \text{ m/s}^2$ (a deceleration of 2 m/s^2).

Velocity-time graphs

- 6 A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s over 6 seconds, then travels at constant velocity for a further 4 seconds. The velocity-time graph is shown. Find the total distance traveled, using the area under the graph.
Distance in first phase (trapezoid): $\frac{1}{2}(2 + 14)(6) = 48$ m. Distance in second phase (rectangle): $14 \times 4 = 56$ m. Total: $48 + 56 = 104$ m.
- 7 A particle decelerates uniformly from $v = 20$ m/s to rest over 5 seconds, giving a velocity-time graph that is a straight line from $(0, 20)$ to $(5, 0)$. Find the distance traveled, using the area under the graph.
The graph forms a right triangle with base 5 and height 20. Distance = $\frac{1}{2}(5)(20) = 50$ m.

Motion under gravity

- 8 A stone is dropped from a height of 45 meters. Using $g = 9.8 \text{ m/s}^2$, find the time taken to reach the ground, and its speed on impact.
 $s = \frac{1}{2}gt^2$: $45 = \frac{1}{2}(9.8)t^2$, so $t^2 = \frac{90}{9.8} \approx 9.18$, giving $t \approx 3.03 \text{ s}$. $v = gt = 9.8(3.03) \approx 29.7 \text{ m/s}$.
- 9 A ball is thrown vertically upward with initial speed $u = 24.5 \text{ m/s}$. Using $g = 9.8 \text{ m/s}^2$, find the total time taken to return to its starting point.
 Time to reach the highest point: $t = \frac{u}{g} = \frac{24.5}{9.8} = 2.5 \text{ s}$. By symmetry, the total time to return to the starting point is $2 \times 2.5 = 5 \text{ s}$.

Using calculus for variable acceleration

- 10 A particle has velocity $v = 3t^2 - 2t \text{ (m/s)}$. Find its displacement from $t = 0$ to $t = 3$, and its acceleration at $t = 3$.
 Displacement: $s = \int_0^3 (3t^2 - 2t) dt = [t^3 - t^2]_0^3 = 27 - 9 = 18 \text{ m}$. Acceleration: $a = \frac{dv}{dt} = 6t - 2$; at $t = 3$: $a = 16 \text{ m/s}^2$.
- 11 A particle has velocity $v = 4t - t^2 \text{ (m/s)}$. Find its displacement from $t = 0$ to $t = 2$, and its acceleration at $t = 2$.
 Displacement: $s = \int_0^2 (4t - t^2) dt = \left[2t^2 - \frac{t^3}{3}\right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3} \approx 5.33 \text{ m}$. Acceleration: $a = \frac{dv}{dt} = 4 - 2t$; at $t = 2$: $a = 0 \text{ m/s}^2$.

Finding when a particle is at rest or changes direction

- 12 A particle has velocity $v = t^2 - 5t + 6 \text{ (m/s)}$. Find the times at which the particle is instantaneously at rest.
 $t^2 - 5t + 6 = 0$, so $(t - 2)(t - 3) = 0$, giving $t = 2$ or $t = 3$ seconds.
- 13 A particle has velocity $v = t^2 - 7t + 10 \text{ (m/s)}$. Find the times at which the particle is instantaneously at rest.
 $t^2 - 7t + 10 = 0$, so $(t - 2)(t - 5) = 0$, giving $t = 2$ or $t = 5$ seconds.

Relative and combined motion

- 14 Two cars start 300 meters apart on a straight road and travel toward each other, one at 18 m/s and the other at 12 m/s, both at constant speed. Find the time until they meet.
 Combined closing speed: $18 + 12 = 30 \text{ m/s}$. Time = $\frac{300}{30} = 10$ seconds.
- 15 Two cars start at the same point on a straight road, one traveling at 25 m/s. The second car starts 4 seconds later, traveling at 30 m/s. Find how long after the first car starts the second car catches up.
 Let t be the time (seconds) after the first car starts. First car's position: $25t$. Second car's position (starting 4 s later): $30(t - 4)$. Setting equal: $25t = 30(t - 4) = 30t - 120$, so $5t = 120$, giving $t = 24$ seconds.

A well-designed word problem: a train between two stations

- 16** This question has two parts. A train starts from rest at station A and accelerates uniformly at 0.5 m/s^2 until it reaches a speed of 20 m/s . It then travels at this constant speed before decelerating uniformly at 1 m/s^2 to come to rest at station B. The total distance between the stations is 2400 meters . (a) Find the time and distance covered during the acceleration phase. (b) Find the total journey time between the two stations.

(a) Acceleration phase: $t_1 = \frac{v - u}{a} = \frac{20}{0.5} = 40 \text{ s}$. Distance: $s_1 = \frac{1}{2}(0 + 20)(40) = 400 \text{ m}$. (b) Deceleration phase: $t_3 = \frac{20}{1} = 20 \text{ s}$, distance $s_3 = \frac{1}{2}(20 + 0)(20) = 200 \text{ m}$. Constant-speed distance: $2400 - 400 - 200 = 1800 \text{ m}$, taking $t_2 = \frac{1800}{20} = 90 \text{ s}$. Total journey time: $40 + 90 + 20 = 150$ seconds.